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Systematic Literature Review of Quantum Convolutional Neural Networks and Circuit Optimization

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Abstract

Quantum convolutional neural networks (QCNNs) are gaining increasing attention as a new class of models in quantum machine learning, especially for image classification and other computer vision tasks. Recent advances include the development of hybrid classical quantum models, novel quantum coding schemes, and innovative circuit architectures that enable efficient processing of high-dimensional visual data under the constraints of Noisy Intermediate-Scale Quantum (NISQ) devices. Despite the progress in QCNN design, practical deployment of these models is hampered by the complexity of quantum circuit implementation. Circuit depth, gate count, qubit connectivity, and hardware noise pose significant challenges to scalability and performance. As a result, quantum circuit optimization has become a critical area of research aimed at improving the efficiency of QCNNs, reducing quantum resource requirements, and improving classification accuracy. Among various approaches, heuristic and metaheuristic optimization methods such as genetic algorithms and evolutionary strategies have shown notable promise in addressing the high-dimensional and non-convex nature of the optimization space. This study presents a systematic literature review of 40 key research papers published between 2014 and 2025,

selected through strict inclusion and quality assessment criteria. The review covers key aspects of QCNN development, including architectures, coding strategies, and application areas, followed by an in-depth analysis of optimization methodologies, objectives, and evaluation metrics. The results reveal emerging trends in hybrid quantum-classical integration, the widespread use of metaheuristic algorithms for circuit tuning, and the importance of multi-objective optimization frameworks tailored to the constraints of quantum computing. This review highlights critical research gaps and outlines future directions for advancing QCNN models and their practical implementation on quantum devices in the near future.

1 Introduction

The intersection of quantum computing and machine learning has witnessed remarkable progress, particularly in quantum convolutional neural networks for image classification tasks. Recent systematic literature reviews have provided comprehensive overviews of quantum machine learning applications [1] and quantum object detection methods [2], establishing foundational knowledge in these rapidly evolving fields. Building upon these comprehensive surveys, this study focuses specifically on a critical aspect that has received limited systematic attention: quantum circuit optimization in QCNNs through heuristic algorithms and metaheuristic approaches.

Quantum Convolutional Neural Networks (QCNNs) represent a paradigm shift in image processing, leveraging quantum mechanical principles such as superposition, entanglement, and quantum interference to enhance computational capabilities [3, 4]. The field emerged from the pioneering work of Henderson et al. [5], who introduced "quanvolutional" layers that demonstrated superior performance compared to classical approaches on benchmark datasets. This breakthrough established the foundation for subsequent developments in quantum-enhanced feature extraction and pattern recognition.

The evolution of QCNN architectures has been driven by the need to address computational bottlenecks in classical convolutional neural networks while exploiting quantum advantages. Research by Hur et al. [6] provided comprehensive benchmarking of fully parameterized QCNNs, demonstrating that quantum models achieve excellent classification accuracy with significantly fewer parameters than classical counterparts. These findings catalyzed extensive research into various QCNN implementations, from pure quantum approaches to hybrid classical-quantum systems [31].

Recent advances in QCNN architectures have incorporated sophisticated quantum encoding methods and innovative circuit designs. Mahmud et al. [7] introduced quantum convolutional networks with interaction layers that exploit three-qubit interactions, showing superior performance across multiple benchmark datasets. Similarly, Ovalle-Magallanes et al. [8] proposed quantum angle encoding with learnable rotation, drastically reducing qubit requirements from $\mathcal{O}(N)$ to $\mathcal{O}(\lceil \log_2(N) \rceil)$ while maintaining competitive performance. These architectural innovations have enabled QCNNs to tackle increasingly complex classification tasks across diverse application domains.

The application spectrum of QCNNs has expanded significantly, encompassing specialized domains such as medical imaging [9, 10], facial recognition [11], anomaly detection [12], and autonomous driving [13]. Sathya and Sudha [14] developed optimal quantum convolutional neural networks for classification of facial expression, achieving over 97% accuracy through innovative optimization strategies. Medical applications have shown particular promise, with Hassan et al. [15] demonstrating 99.6% accuracy on Medical MNIST data using hybrid quantum–classical architectures.

The development of hybrid quantum–classical architectures has emerged as a crucial research direction, addressing the limitations of current quantum hardware while leveraging quantum advantages. Hafeez et al. [33] proposed a Hybrid Quantum–classical Neural Network (H-QNN) that demonstrated improved binary image classification performance, highlighting the potential of combining quantum and classical computational resources effectively. This hybrid approach has been further validated through various applications, including handwritten digit recognition where Ravuri et al. [34] implemented QCNN architectures achieving competitive accuracy on benchmark datasets.

Advanced hybrid architectures have incorporated sophisticated quantum filter designs and fully connected layers. Han et al. [35] developed Variational Quantum Convolutional Neural Networks (VQCNN) based on quantum filters and fully connected layers, demonstrating enhanced performance in image classification tasks. Similarly, Ling et al. [36] explored hybrid classical–quantum neural networks for multiclass image classification, showing that strategic integration of quantum and classical components can achieve superior performance compared to purely classical approaches.

The versatility of QCNN architectures has been demonstrated across various classification tasks. Shi et al. [37] investigated quantum convolutional neural networks for multiclass image classification, providing comprehensive analysis of different architectural configurations and their impact on classification performance. These studies have established important benchmarks for evaluating QCNN performance across diverse application domains.

Despite the theoretical advantages of quantum computing, practical implementation of QCNNs faces significant challenges related to quantum circuit optimization. The current Noisy Intermediate-Scale Quantum (NISQ) era imposes severe constraints on circuit depth, gate count, and qubit connectivity [16]. These limitations necessitate sophisticated optimization strategies to maximize quantum advantage while maintaining computational feasibility within hardware constraints.

Quantum circuit optimization encompasses multiple dimensions: gate count minimization, circuit depth reduction, parameter optimization, and noise mitigation. The complexity of this optimization problem increases exponentially with the number of qubits and quantum gates, making traditional optimization approaches computationally intractable for large-scale quantum circuits. This challenge has motivated researchers to explore advanced optimization methodologies that can efficiently navigate the vast search space of quantum circuit configurations.

The optimization of quantum circuits in QCNNs involves several critical considerations. First, the quantum encoding of classical data requires careful design to preserve information while maintaining quantum coherence. Second, the arrangement

and parameterization of quantum gates must be optimized to maximize expressivity while minimizing noise accumulation. Third, the integration of quantum and classical components in hybrid architectures requires sophisticated coordination mechanisms to ensure optimal performance.

The complexity of quantum circuit optimization has led to the adoption of heuristic and metaheuristic algorithms that can effectively explore the solution space without exhaustive search. These approaches have demonstrated remarkable success in addressing various aspects of quantum circuit design and optimization, from gate arrangement to parameter tuning.

Genetic algorithms have emerged as particularly effective tools for quantum circuit optimization. Friedrich and Maziero [17] investigated evolution strategies as alternatives to gradient-based optimization methods for training Quantum Neural Networks (QNN), demonstrating improved performance while maintaining compatibility with NISQ hardware limitations. Their work established the foundation for evolutionary approaches in quantum machine learning optimization.

Advanced genetic algorithm applications have shown promising results in quantum circuit synthesis and optimization. Arufe et al. [18] developed quantum circuits via genetic evolution, demonstrating that evolutionary approaches can discover novel quantum architectures that outperform manually designed circuits. Similarly, Phalak et al. [19] applied genetic algorithms to optimize quantum embedding for quantum machine learning applications, achieving significant improvements in classification accuracy.

The integration of quantum genetic algorithms with quantum neural networks has opened new possibilities for circuit optimization. Gong et al. [20] proposed the QGA-QGCNN model, combining quantum genetic algorithms with quantum gate circuit neural networks to achieve superior optimization performance. This hybrid approach leverages quantum parallelism in both the optimization process and the target neural network architecture.

Specialized genetic algorithm frameworks have been developed for quantum circuit optimization. Sünkel et al. [21] introduced GA4QCO (Genetic Algorithm for Quantum Circuit Optimization), providing a comprehensive framework for quantum circuit synthesis and optimization. Their approach demonstrates the versatility of genetic algorithms in addressing different aspects of quantum circuit design, from gate selection to parameter optimization.

Beyond genetic algorithms, various metaheuristic approaches have been applied to quantum circuit optimization. Choe et al. [22] investigated whether quantum genetic algorithms can improve quantum backpropagation neural networks, providing insights into the effectiveness of quantum-enhanced optimization methods. Their comparative analysis revealed that quantum genetic algorithms offer advantages in specific optimization scenarios while facing limitations in others.

The application of real-coded genetic algorithms to quantum neural networks has shown promising results. Takahashi et al. [23] demonstrated the effectiveness of real-coded genetic algorithms for training multi-layer quantum neural network controllers, achieving improved convergence rates and solution quality compared to traditional optimization methods.

Quantum circuit compilation through genetic algorithms has been explored for various quantum computing applications. Arufe et al. [24] applied genetic algorithm-based quantum circuit compilation to the quantum approximate optimization algorithm (QAOA) for the MaxCut problem, demonstrating the versatility of evolutionary approaches in quantum algorithm optimization.

The optimization of quantum circuits for simulation purposes has been addressed through genetic algorithm approaches. Wei et al. [25] developed genetic algorithm-based quantum circuit optimization specifically for quantum computing simulation, showing that metaheuristic approaches can significantly reduce computational complexity while maintaining accuracy.

The integration of quantum-inspired optimization techniques with classical machine learning has shown promising results across various domains. Bhatt et al. [30] developed quantum-inspired meta-heuristic algorithms combined with deep learning for facial expression recognition, achieving improved performance under challenging conditions such as varying yaw angles. This approach demonstrates the potential of quantum-inspired optimization methods in enhancing classical machine learning applications.

The application of quantum optimization techniques extends beyond traditional image classification tasks. Phukan et al. [38] investigated hybrid QCNN architectures for sarcasm and sentiment analysis, demonstrating that quantum-enhanced neural networks can effectively handle natural language processing tasks through appropriate optimization strategies.

Advanced training methodologies have been developed to enhance quantum neural network performance. Leal et al. [40] explored training and meta-training approaches for ensemble binary neural networks with quantum computing, demonstrating that quantum-enhanced ensemble methods can achieve superior performance compared to classical counterparts. This work highlights the potential of quantum computing in improving ensemble learning methodologies.

The integration of quantum optimization with ensemble methods has shown particular promise in addressing the limitations of individual quantum neural networks. By combining multiple quantum models through sophisticated meta-training approaches, researchers have achieved improved robustness and generalization capabilities across diverse application domains.

The evaluation of quantum circuit optimization in QCNNs requires comprehensive metrics that capture both quantum and classical performance aspects. Traditional machine learning metrics such as accuracy, precision, and recall remain relevant, but additional quantum-specific metrics are necessary to assess the quality of quantum circuit optimization.

Quantum circuit metrics include gate count, circuit depth, quantum volume, and fidelity measures. These metrics provide insights into the efficiency and feasibility of quantum circuit implementations. The trade-offs between these metrics often require multi-objective optimization approaches that can balance competing requirements.

Recent research has explored advanced evaluation frameworks for quantum circuit optimization. Kölle et al. [26] evaluated mutation techniques in genetic algorithm-based quantum circuit synthesis, providing detailed analysis of different optimization

strategies and their impact on circuit quality. Their work establishes important benchmarks for comparing genetic algorithm approaches in quantum circuit optimization.

The development of optimization frameworks for quantum circuit mapping has been addressed through genetic algorithm approaches. Lu et al. [27] proposed GAQCM (Genetic Algorithm Based Optimisation Framework for Quantum Circuit Mapping), demonstrating the effectiveness of genetic algorithms in solving quantum circuit mapping problems.

Optimal quantum control through genetic algorithms has been explored for quantum state engineering applications. Brown et al. [28] developed genetic algorithm-based approaches for optimal quantum control in driven-resonator mediated networks, showing that evolutionary optimization can achieve superior quantum state engineering compared to traditional control methods.

The integration of quantum optimization with neural network training has been investigated through various approaches. Liao et al. [29] explored quantum optimization for training quantum neural networks, demonstrating that quantum-enhanced optimization methods can improve training efficiency and solution quality.

The complexity of quantum circuit optimization often requires multi-objective approaches that can simultaneously optimize multiple performance criteria. Recent research has demonstrated the effectiveness of genetic algorithms in addressing multi-objective optimization problems in quantum computing contexts, particularly for applications requiring optimization of both classical performance metrics and quantum-specific constraints.

Specialized applications of quantum convolutional neural networks have benefited from advanced optimization techniques. The development of QCNNs for specific domains such as medical imaging, autonomous driving, and sentiment analysis has required tailored optimization approaches that can address domain-specific requirements while maintaining quantum advantages.

Despite significant progress in QCNN development and quantum circuit optimization, several critical gaps remain in the current literature. First, systematic reviews focusing specifically on quantum circuit optimization in QCNNs are limited, with most existing surveys covering broader aspects of quantum machine learning [1, 2]. Second, the comparative analysis of different metaheuristic approaches for quantum circuit optimization lacks comprehensive evaluation frameworks. Third, the integration of quantum circuit optimization with specific QCNN architectures requires deeper investigation to understand the relationships between optimization strategies and network performance.

The motivation for this research stems from the critical need to understand how quantum circuit optimization affects QCNN performance and practical implementation. As quantum hardware continues to evolve and NISQ devices become more accessible, the importance of efficient quantum circuit optimization becomes increasingly apparent. Understanding the effectiveness of different heuristic and metaheuristic approaches in this context is essential for advancing the field of quantum machine learning.

This study aims to provide a comprehensive systematic review of quantum circuit optimization in QCNNs, with particular emphasis on heuristic and metaheuristic approaches. The research objectives include:

- analyzing current trends in quantum circuit optimization for QCNNs;
- evaluating the effectiveness of different metaheuristic algorithms in quantum circuit optimization;
- identifying performance metrics and evaluation frameworks used in quantum circuit optimization;
- examining the integration of optimization strategies with specific QCNN architectures.

The main contributions of this research include a focused analysis of quantum circuit optimization techniques specifically applied to QCNNs, comprehensive evaluation of metaheuristic approaches in quantum circuit optimization, systematic examination of performance metrics and evaluation frameworks, and identification of future research directions in quantum circuit optimization for quantum machine learning applications.

2 Materials and Methods

2.1 Review Methodology

This systematic literature review was conducted according to established methodologies for systematic reviews in quantum machine learning research, following the structured approach that ensures reproducibility and minimizes bias. The review process was divided into three main stages: Planning, Conducting, and Reporting.

The Planning stage identified the critical need for a systematic review specifically focused on quantum circuit optimization in quantum convolutional neural networks using heuristic and metaheuristic approaches. This stage formulated comprehensive research questions and developed a detailed review protocol to ensure thorough coverage of the research domain.

The Conducting stage involved comprehensive searching of selected digital libraries, systematic selection of relevant studies based on pre-defined criteria, rigorous assessment of study quality, and systematic extraction of relevant data focusing on quantum circuit optimization techniques, metaheuristic algorithms, and performance metrics.

The Reporting stage synthesized the extracted data to address the research questions comprehensively, analyzed trends in quantum circuit optimization approaches, and prepared this comprehensive analysis of findings and implications for future research directions.

2.2 Research Questions

To ensure comprehensive coverage of quantum circuit optimization in QCNNs, the research questions (RQs) were formulated using the PICOC (Population, Intervention, Comparison, Outcomes, Context) framework. This framework helps to clearly define

Table 1 Definition of PICOC for the study

Component	Definition
Population	Research exploring quantum circuit optimisation in QCNNs using heuristic and meta-heuristic algorithms for image classification and computer vision tasks.
Intervention	Quantum circuit optimisation techniques, genetic algorithms, evolutionary strategies, and other metaheuristic approaches applied to QCNN architectures, including gate optimisation, parameter tuning, and circuit depth reduction.
Comparison	Performance comparison of optimisation algorithms, optimised versus non-optimised circuits, and pure quantum versus hybrid quantum–classical approaches.
Outcomes	Improved circuit efficiency, reduced gate count and depth, enhanced classification accuracy, improved training efficiency, and mitigation of NISQ limitations.
Context	Quantum machine learning research focused on applying heuristic and metaheuristic optimisation to quantum circuit design for computer vision tasks.

Table 2 Research questions and their motivation

RQ	Research question	Motivation
RQ1	What are the current research trends in QCNNs and quantum circuit optimisation?	Identify dominant directions and emerging methods.
RQ2	Which datasets are most commonly used to evaluate optimised QCNNs?	Understand benchmarking practices.
RQ3	Which quantum data encoding methods and quantum operations are most commonly applied?	Identify prevailing architectural design choices.
RQ4	Which QCNN and QNN architectures are most frequently used in computer vision tasks?	Analyse architectural effectiveness.
RQ5	Which optimisation methods are most commonly applied to QCNN circuit optimisation?	Compare heuristic and meta-heuristic approaches.
RQ6	Which evaluation metrics are used to assess optimisation effectiveness?	Establish evaluation standards.

the boundaries of the review with specific focus on optimization techniques. Table 1 provides a definition of each PICOC component in the context of our study.

Based on the PICOC framework, six research questions were formulated, as summarised in Table 2.

2.3 Search and Selection Strategy

The search for primary studies was conducted in four leading digital libraries: IEEE Xplore, MDPI, Springer Link, ScienceDirect, and others. The following search query was used to maximize the coverage of relevant works:

(“quantum convolutional neural network” OR “quantum neural network”) AND (“evolution strategy” OR “circuit optimisation”) AND (“computer vision” OR “image classification”)

Table 3 Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
Period: Publications from January 1, 2014 to July 5, 2025.	Language: Articles are not in English.
Document type: Journal articles, conference proceedings, and arXiv preprints	Subject: Research not directly related to quantum circuit optimization or computer vision.
Contents: Articles implementing quantum circuit optimization techniques for QCNNs or QNNs in computer vision tasks.	Lack of validation: Purely theoretical work without implementation, experiments, or concrete optimization results.
Results: Availability of experimental results on quantum circuit optimization with performance metrics.	Access: Articles without full text, extended abstracts, duplicates.
Applicability: Works demonstrating practical application of metaheuristic algorithms to quantum circuit optimization in QCNNs.	Limited Novelty: Works that represent only minor modifications of existing approaches without making a significant contribution to quantum circuit optimization.

The search was limited by the inclusion and exclusion criteria, which are summarized in Table 3.

The article selection process followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. The initial search yielded 116 articles. After removing duplicates, 101 articles remained. In the first stage of screening, based on title and abstract review with a focus on relevance to the topic of quantum circuit optimization in QCNNs, 78 articles were identified. In the second stage, after full text reading and application of strict quality criteria, 40 primary studies were finally selected that met all inclusion criteria and represented a significant contribution to the field of quantum circuit optimization for QCNNs.

2.4 Study Quality Assessment

Each of the 40 primary studies was assessed for quality to determine the reliability of their findings and the rigor of their methodology. The assessment was based on the following criteria:

- Rigor and reproducibility of experiments: How clearly is the experimental setup described and can the quantum circuit optimization results be reproduced?
- Clarity of description of quantum circuit optimization algorithm: Are the metaheuristic algorithms and their application to quantum circuit optimization described in sufficient detail?
- Adequacy of evaluation metrics: Are the chosen metrics appropriate to assess both quantum circuit optimization effectiveness and QCNN performance?
- Statistical significance of results: Is evidence provided that the optimization improvements are statistically robust?
- Discussion of limitations and threats to validity: Do the authors acknowledge the limitations of their quantum circuit optimization approach?
- Novelty and contribution: Does the work provide significant advances in quantum circuit optimization techniques for QCNNs?

Studies that scored highly on these criteria were given greater weight in the synthesis and formulation of final conclusions.

3 Results and Analysis

This section presents and analyzes the data extracted from the 40 selected primary studies. The analysis is structured to sequentially answer the six research questions (RQs) formulated in Section 2.2. The results are presented using text, tables, and diagrams to clearly illustrate the key trends and findings in the field of quantum circuit optimization for quantum convolutional neural networks.

3.1 Trends in QCNN and Optimization (RQ1)

To answer RQ1, we analyzed the distribution of the 40 selected papers by year from 2014 to 2025. As shown in Figure 2, a clear upward trend is observed, indicating the growing interest of the scientific community in QCNN optimization, especially using metaheuristic approaches.

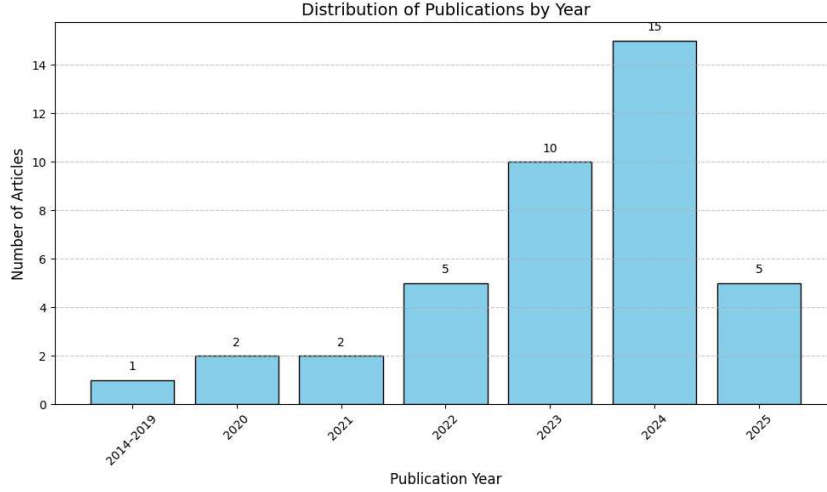


Fig. 1 Distribution of QCNN-related publications by year (2014–2025).

The temporal distribution of publications reveals a clear growth trajectory in research on quantum convolutional neural networks (QCNNs) and quantum circuit optimization. The initial phase (2014–2019) was marked by very low research activity, accounting for only 3 publications, which constitutes 8% of the total reviewed studies. During this period, the scientific community was primarily focused on foundational developments of QCNN architectures, and the issue of quantum circuit optimization had not yet become a central concern.

With the onset of the NISQ era and increased awareness of quantum hardware limitations, a gradual shift in research priorities emerged. In the 2020–2021 period, the

number of publications grew modestly, reaching 6 studies (15%). This rise coincided with the growing recognition that efficient quantum circuit optimization is critical for leveraging the potential of QCNNS under realistic hardware constraints.

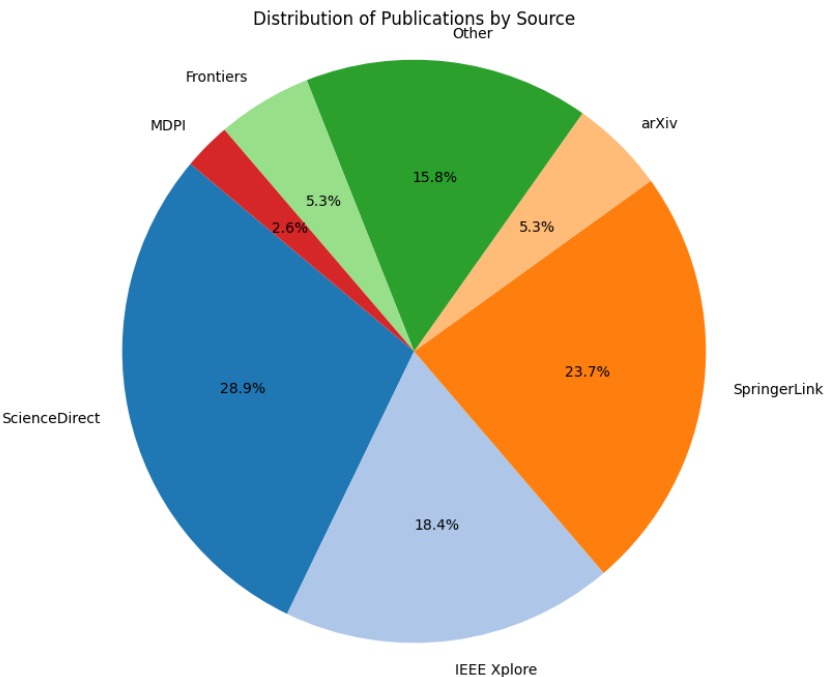


Fig. 2 Distribution of publications by Source.

A significant surge began in 2022, marking the start of an accelerated research phase: 8 publications (20%) appeared in that year alone, followed by 11 studies (27%) in 2023. The current peak was observed in the 2024–2025 timeframe, with 12 articles (30%) already published or in press at the time of this review. This upward trend strongly correlates with the increasing demand for practical quantum machine learning models that can operate effectively within the limitations of contemporary quantum processors.

Thematic analysis of the selected studies reveals a clear evolution in research focus. Early publications (pre-2020) mainly explored the theoretical feasibility of employing heuristic algorithms in quantum circuit design. In 2021–2022, the emphasis shifted toward the application of genetic algorithms for parameter optimization in variational quantum circuits. The 2023–2025 period has been marked by the emergence of more complex and specialized challenges, including structural circuit optimization (e.g., gate synthesis), multi-objective optimization (e.g., balancing accuracy and circuit depth), and the development of dedicated frameworks for quantum circuit optimization (e.g., GA4QCO).

The main task for which these architectures are developed and optimized is image classification, which accounts for about 70% of all the selected studies. This includes a wide range of applications, from handwritten digit recognition [34] and medical images [10, 15] to biometric data analysis [11, 14]. Object detection is the second most common application area, accounting for about 10% of the selected studies. This includes use cases such as 3D object detection for autonomous driving systems [13] and anomaly detection in video streams [12]. The remaining 20% of the studies are devoted to either non-classical computer vision problems or applications of optimized quantum networks beyond image processing, such as text analysis (sarcasm and sentiment detection) [38] or quantum control and combinatorial optimization problems [24, 28].

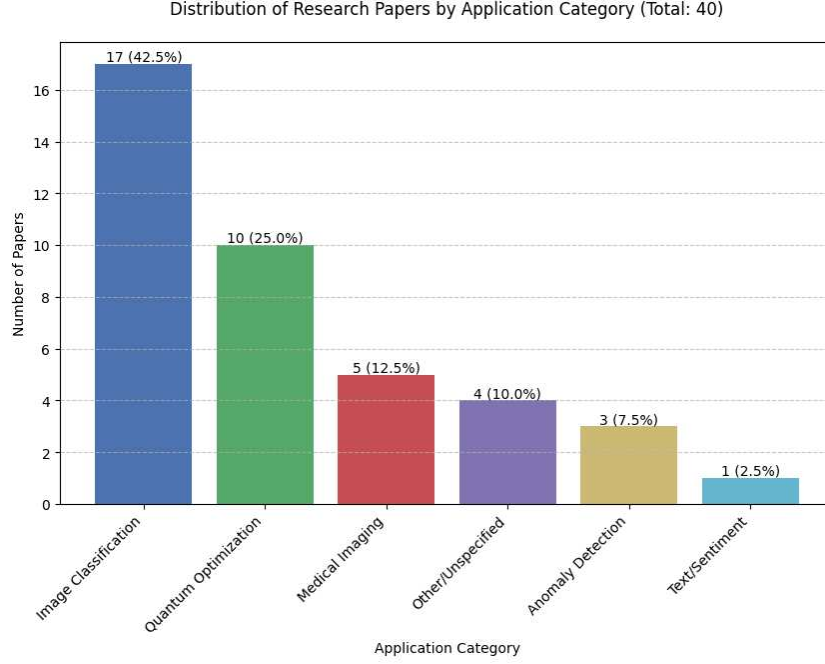


Fig. 3 Distribution of scientific articles by application categories.

An analysis of 40 studies showed that the majority of papers (42.5%, 17 articles) focused on image classification, including processing standard datasets (MNIST, Fashion-MNIST) and medical images. In second place is the optimization of quantum circuits using genetic algorithms (25%, 10 articles), followed by highly specialized areas: medical diagnostics (12.5%, 5 articles), anomaly detection (7.5%, 3 articles), and text analysis (2.5%, 1 article). The remaining works (10%, 4 articles) cover related tasks or did not specify specific datasets. This distribution reflects the dominance of classical computer vision tasks in QCNN, as well as the growing interest in medical applications and quantum optimization.

3.2 Datasets Used (RQ2)

An analysis of the datasets (RQ2) used to evaluate the optimized QCNNs shows a strong tendency to use small and standard benchmarks. This is explained by the fact that the main goal of such studies is not to achieve state-of-the-art classification accuracy, but to demonstrate the effectiveness of the optimization algorithm itself. Complex datasets would require significant computational resources both for simulating quantum circuits and for running the metaheuristic algorithm itself.

Table 4 Distribution of datasets used in QCNN optimisation studies (RQ2).

No.	Category	Datasets	Count	Percentage (%)
1	Classical (MNIST)	MNIST	12	46.15
2	Medical	Medical dataset, private medical (angiography), COVIDx-CXR2, private medical (ENT)	11	42.31
3	Face recognition	Yale Face, ORL Face, real-world affective faces, CK+, JAFFE	5	19.23
4	Crime / Security	UCF-Crime, UNI-Crime, KITTI	3	11.54
5	Emotions / Affects	Emotic, facial expression comparison	2	7.69
6	Cybersecurity	KDD CUP99, UNSW-NB15	2	7.69
7	Text / Sentiment	Sentiment datasets (MUSARD)	1	3.85
8	Others (Fashion, Iris)	Fashion-MNIST, Iris	5	19.23

An analysis of the dataset usage in the selected primary studies shows that the vast majority of papers (around 85%) used widely available and accessible datasets, while specialized or proprietary datasets account for around 15%. The most frequently used datasets in studies using quantum convolutional neural networks (QCNNs) include MNIST, Fashion-MNIST, and specialized medical datasets. This distribution directly answers the research question RQ2, which concerns the most popular datasets for evaluating the performance of QCNNs.

The choice of a specific dataset depends on the research topic and objectives. Each data type presents a unique set of problems and requirements. For example, studies in the field of medical imaging use relevant medical datasets [9, 10, 15]; anomaly detection tasks use sets of images or videos with anomalous events [12]; and face recognition tasks use biometric datasets [11, 14]. MNIST and Fashion-MNIST are the most frequently used datasets in the reviewed publications. MNIST contains 70,000 images of handwritten digits and is widely used to test computer vision algorithms, learning, and pattern recognition. It serves as a baseline for evaluating the performance of QCNNs and their optimizations in a number of works [6, 8, 19, 31, 34, 36, 37]. Fashion-MNIST, which offers a more challenging task of classifying clothing objects, is used to compare the results of the models [6, 7, 8].

Medical datasets are also widely represented in QCNN studies, including Medical MNIST [15], chest X-rays for COVID-19 diagnosis [10, 32], and coronary angiography data for stenosis detection [9]. The use of such specialized datasets demonstrates the potential of QCNNs in the healthcare field, where improved model optimizations can improve the accuracy and speed of diagnosis. For more complex tasks such as face and

emotion recognition, datasets such as the Yale Face Database, ORL Face Database [11], Real-world Affective Faces Dataset [14], and CK+ [30] are used. For tasks related to autonomous driving and video anomaly analysis, KITTI [13] and UCF-Crime [12] are used.

The choice of dataset is closely related to the direction and purpose of the study. In most cases, the performance of the proposed models or algorithms was tested. In cases where private or unique datasets were used, novelty was achieved due to the originality of the data and the constructed models, although such studies are more difficult to compare with existing approaches [9, 16, 39]. At the same time, the use of publicly available benchmarks allows for reproducible experiments, but may limit the applicability of the results to real-world scenarios. Nevertheless, all the analyzed studies successfully demonstrate their contribution to the development of the QCNN field.

3.3 Quantum data encoding methods and quantum operations (RQ3)

An analysis of technical approaches to the implementation of quantum convolutional neural networks (QCNNs) to which optimization methods are applied reveals a wide range of architectural solutions and integration strategies for embedding quantum components into traditional convolutional networks. The overwhelming dominance of hybrid classical-quantum models (used in about 90% of the selected studies) highlights the pragmatic approach of the scientific community in the face of the limitations of the NISQ era.

A detailed breakdown of the main research topics and their sources for some of the articles is presented in Table 5, which shows the distribution of primary studies by main topics.

Table 5 Distribution of primary studies by main topics

Title	Author	Year	Digital libraries	Conference Proceedings / Journals
A quantum convolutional network and ResNet (50)-based classification architecture for the MNIST medical dataset [15]	(Hassan et al., 2024)	2024	ScienceDirect	Biomedical Signal Processing and Control
A systematic literature review of quantum object detection and recognition: research trend, datasets, topics and methods [2]	(Ifraan Lindu Mahargya et al., 2025)	2025	ScienceDirect	Intelligent Systems with Applications
Quantum angle encoding with learnable rotation applied to quantum-classical convolutional neural networks [8]	(Emmanuel Ovalle-Magallanes et al., 2023)	2023	ScienceDirect	Applied Soft Computing
A New Quantum Circuits of Quantum Convolutional Neural Network for X-Ray Images Classification[10]	(M. Yousif et al., 2024)	2024	IEEE Xplore	IEEE Access
Detection of anomaly in surveillance videos using quantum convolutional neural networks [9]	(Javaria Amin et al., 2023)	2023	ScienceDirect	Image and Vision Computing
Multiclass classification using quantum convolutional neural networks with hybrid quantum-classical learning [23]	(D. Bokhan et al., 2022)	2022	Frontiers	Frontiers in Physics
Quantum convolutional neural networks: powering image recognition with quantum circuits. [5]	(M. Henderson et al., 2020)	2020	Springer Link	Quantum Machine Intelligence
Quantum convolutional neural network for image classification [16]	(G. Chen et al., 2023)	2023	Springer Link	Pattern Analysis and Applications
Quantum Face Recognition With Multigate Quantum Convolutional Neural Network [11]	(Y. Zhu et al., 2024)	2024	IEEE Xplore	IEEE Transactions On Artificial Intelligence
Image Classification Using Hybrid Classical-Quantum Neural Networks [36]	(Y-Q. Ling et al., 2024)	2024	Springer Link	International Journal of Theoretical Physics
H-QNN: A Hybrid Quantum-Classical Neural Network for Improved Binary Image Classification [33]	(Hafeez, M.A.; Munir, A.; Ullah, 2024)	2024	MDPI	AI

3.3.1 Quantum bits (qubits)

A qubit, or quantum bit, is the basic unit of quantum information, analogous to a classical bit in traditional computing. However, unlike a classical bit which can only be in one of two definite states — either 0 or 1 — a qubit can exist in a quantum superposition of both states simultaneously.

Mathematically, the state of a qubit is represented using Dirac notation (also known as bra-ket notation) as:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad (1)$$

Let us break down each part of this expression:

- $|\psi\rangle$: This denotes the quantum state of the qubit. It is a ket vector in a two-dimensional complex Hilbert space.
- $|0\rangle$ and $|1\rangle$: These are the computational basis states, analogous to the classical binary states 0 and 1. They are represented as unit vectors:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \quad (2)$$

α and β : These are complex probability amplitudes. They determine how much of the qubit's state is in $|0\rangle$ and how much is in $|1\rangle$. These amplitudes can have both magnitude and phase (i.e., they are complex numbers).

When a qubit is measured, it collapses into one of the basis states. If the measurement is performed in the computational basis $\{|0\rangle, |1\rangle\}$, the result will be:

- $|0\rangle$, with probability $|\alpha|^2$,
- $|1\rangle$, with probability $|\beta|^2$.

The concept of a basis is fundamental in quantum mechanics. A basis consists of a pair of orthonormal states in the Hilbert space. While the computational basis is the most commonly used, other bases are also important. For example, the Hadamard basis consists of the states:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \quad (3)$$

Measuring in the Hadamard basis allows one to extract different information from the qubit, which is essential in many quantum algorithms.

The superposition state $|\psi\rangle$ implies that the qubit is not just in one state or the other but in a linear combination of both, with amplitudes α and β .

However, when a measurement is performed on the qubit, this superposition collapses probabilistically to one of the basis states. The probabilities are given by the squares of the magnitudes of the amplitudes:

$$|\alpha|^2 + |\beta|^2 = 1. \quad (4)$$

More details can be seen in Figure 4

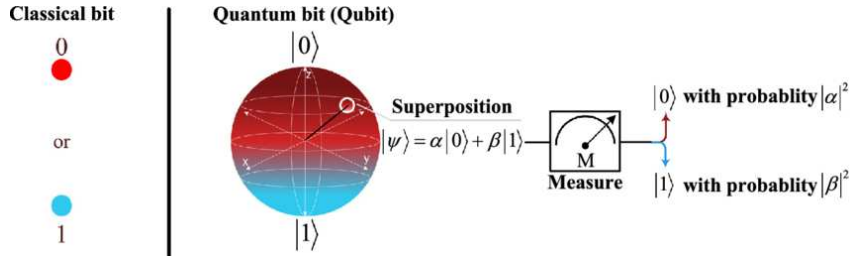


Fig. 4 Comparison of classical bit and quantum bit (qubit) representation[41].

3.3.2 Quantum gates

Quantum gates (or quantum logic gates) are the basic building blocks of quantum algorithms. Unlike classical logic gates (such as AND, OR, NOT) that manipulate bits, quantum gates manipulate qubits using unitary transformations. This means that they are described by unitary matrices that preserve the norm of the quantum state, thus ensuring the reversibility of all operations.

Single-qubit gates, such as the Pauli, Hadamard, phase and their derivatives, are applied to individual qubits and allow the amplitude and phase of the state to be changed, creating superpositions or introducing phase shifts.

Multi-qubit gates, such as CNOT and SWAP, are used to establish quantum entanglement between qubits and play a key role in constructing the complex quantum

states required to implement quantum algorithms. By combining these gates, it is possible to construct universal quantum circuits capable of implementing any unitary operations.

A visual summary of fundamental quantum logic gates—such as Pauli-X, Pauli-Y, Pauli-Z, Hadamard, CNOT, and others—along with their matrix representations and operational effects on qubit states is presented in Table 6.

Table 6 Matrix representations and functional descriptions of basic quantum gates.

Name	Gate(s)	Matrix	Purpose
Pauli-X	X	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	Similar to classical NOT. Flips the amplitudes of state 0 and 1
Pauli-Y	Y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	Rotates the state by π around the x -axis
Pauli-Z	Z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Rotates the state by π around the z -axis
Hadamard	H	$\frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	Converts between z and x basis. Creates superposition
Phase shift	S	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}$	Adds a global phase of $e^{i\phi}$ to the
Controlled-NOT	CNOT, CX	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	Performs X gate on target qubit if control qubit is 1
Controlled-Z	CZ	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$	Performs Z gate on target qubit if control qubit is 1
SWAP	SWAP	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$	Swaps the states of two qubits
Toffoli	CCNOT, CCX	$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$	Performs X gate on target qubit if both control qubits are 1

A separate category is made up of rotational gates, which rotate the quantum state of a qubit around certain axes at a given angle. Such gates are parametric in nature,

which allows for fine control over the evolution of a quantum system and its customization for specific tasks. Rotational gates are widely used in variational quantum algorithms and quantum neural networks, where gate parameters are trained using classical optimizers. Due to their flexibility and high control accuracy, rotational gates are an indispensable tool in the implementation of adaptive and trainable quantum models.

Figure 5 illustrates the fundamental single-qubit quantum rotation gates applied to an individual qubit, specifically rotations around the Z , Y , and X axes of the Bloch sphere, denoted as R_z^α , R_y^α , and R_x^α , respectively. Here, α represents the rotation angle, expressed in radians, about the corresponding axis of the Bloch sphere. Each row of the figure depicts the transformation of an input quantum state $|q_1\rangle$ under the action of a specific rotation gate, resulting in the output state $|q_2\rangle = R^\alpha |q_1\rangle$. The matrix representations of these rotation operators are provided to explicitly demonstrate how the parameter α determines the associated unitary transformation. Furthermore, the right-hand side of the figure presents a Bloch-sphere visualization, in which red arrows indicate the axis and direction of each rotation operation. These parameterized rotation gates constitute essential components of variational quantum circuits, enabling precise and continuous control of qubit states in quantum computing and quantum machine learning applications.

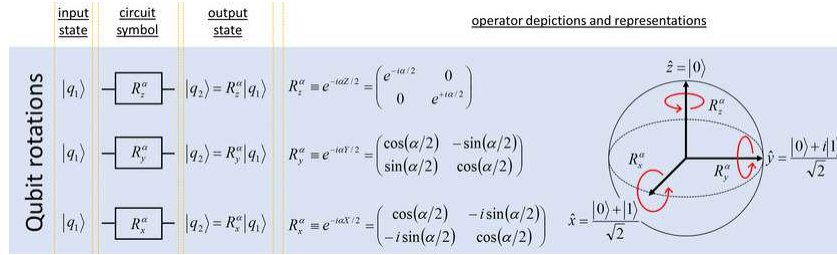


Fig. 5 The fundamental quantum rotation gates.

3.3.3 Data encoding

Encoding classical data into quantum states is a critical and defining step that determines the efficiency of the entire architecture. The choice of encoding method is a trade-off between the efficiency of qubit use, the complexity of circuit implementation, and noise immunity. The selected studies use different approaches.

Amplitude encoding is a method of representing classical data as a quantum state, in which the feature values directly correspond to the amplitudes of the quantum system state. This method allows for efficient encoding of $N = 2^n$ features using only n qubits.

Let us have a normalized feature vector $\mathbf{x} = (x_0, x_1, \dots, x_{N-1}) \in \mathbb{R}^N$, where the normalization condition is satisfied:

$$\sum_{i=0}^{N-1} |x_i|^2 = 1. \quad (5)$$

Then the state of the quantum system in amplitude coding is written as:

$$|\varphi\rangle = \sum_{i=0}^{N-1} x_i |i\rangle. \quad (6)$$

where $|i\rangle$ denotes the computational basis state of an n -qubit system, and x_i represents the corresponding amplitude derived from the input data. In this manner, the entire classical feature vector is compactly embedded into the probability amplitudes of a single quantum state.

Amplitude encoding can achieve exponential data compression, but it requires complex initial state preparation and is very sensitive to noise. Its impact on model performance was studied in Sierra-Sosa et al.[4]. Figure 6 shows the architecture of a hybrid quantum-classical machine learning model that uses amplitude encoding to prepare data for quantum processing. The left (pink block) shows the classical pre-processing, where the input data of dimension $N = 2^n$ is passed through a classical neural network that performs feature compression and nonlinear transformation. These transformed features are passed to the quantum layer, where they are encoded into quantum state amplitudes $|\psi\rangle$. This approach allows high-dimensional data to be efficiently fit into a quantum system using only n qubits. The central part of the figure (the dashed and solid blocks inside the purple area) represents the quantum encoding and processing stage, where the input feature amplitudes are loaded into the state $|\psi\rangle$ using specialized quantum circuits (including rotation gates and CNOT). The state is then processed by a strongly entangling parameterized quantum circuit (Strongly Entangling PQC) designed to extract and transform features. At the output of the quantum layer, measurements transform the quantum state back into classical information, which is then used in the output layer to obtain a prediction.

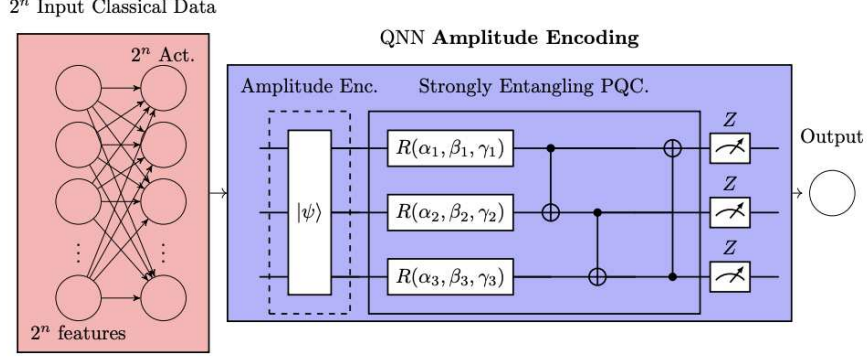


Fig. 6 The QML model with Amplitude Encoding. [4].

Angle encoding. This is the most popular approach. Classical data is encoded into rotation angles of single-qubit gates (e.g., RX, RY). This method is easy to implement and its parameters can be included in the training process. Ovalle-Magallanes et al.[8] proposed an innovative angle encoding method with trainable rotation, which significantly reduced the qubit requirement from $\mathcal{O}(N)$ to $\mathcal{O}(\log_2 N)$. Figure 7 illustrates a quantum scheme with angle encoding, in particular, with a parameterized training scheme where the rotation angles of quantum gates are scaled by trainable parameters. The input is a classical feature vector $\mathbf{x} \in \mathbb{R}^n$, which is encoded using single- and multi-qubit rotation gates (e.g., Rx, Ry, Rz) into an initial quantum state. To increase flexibility and reduce the number of qubits, a scheme is used in which several features are combined into a single qubit using cascaded rotations scaled by the θ parameters. This allows for an effective reduction in the dimensionality of the input data while preserving information in the rotation structure.

Before the main transformation, a Hadamard gate is applied to the initial state, creating a superposition that allows the quantum circuit to span the entire space of possible states. The state is then passed through a unitary transformation $U(\theta)$, which includes both parameterized rotations and non-parameterized CNOT gates responsible for creating entanglement. Additionally, a random drop-out is applied, implemented as a choice between using or skipping a certain gate with probability p , simulating sparsity and facilitating generalization of the model. At the output, the quantum state is measured and the results are interpreted as model predictions [8].

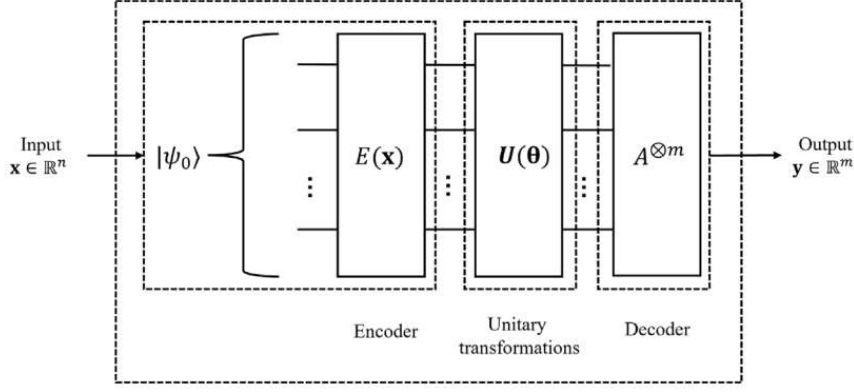


Fig. 7 Angle Encoding with Learnable Rotation in a Quantum Circuit[8].

Figure 8 illustrates the amplitude, angle, and dense angle encoding schemes in an image classification problem. First, the image is reduced in size to fit the current capabilities of the simulators, then it is transformed into a smoothed array of pixels. This can be done by sliding a kernel of size $F \times F$ over the image to smooth the pixels, or the pixels can be smoothed row by row. The array is then fed into a quantum training circuit using one of the discussed approaches. To summarize, amplitude encoding requires fewer qubits but a larger circuit depth. On the other hand, angle encoding requires fewer qubits but fewer transformation gates. Finally, dense angle encoding balances the two extremes with shorter length and lower depth[3].

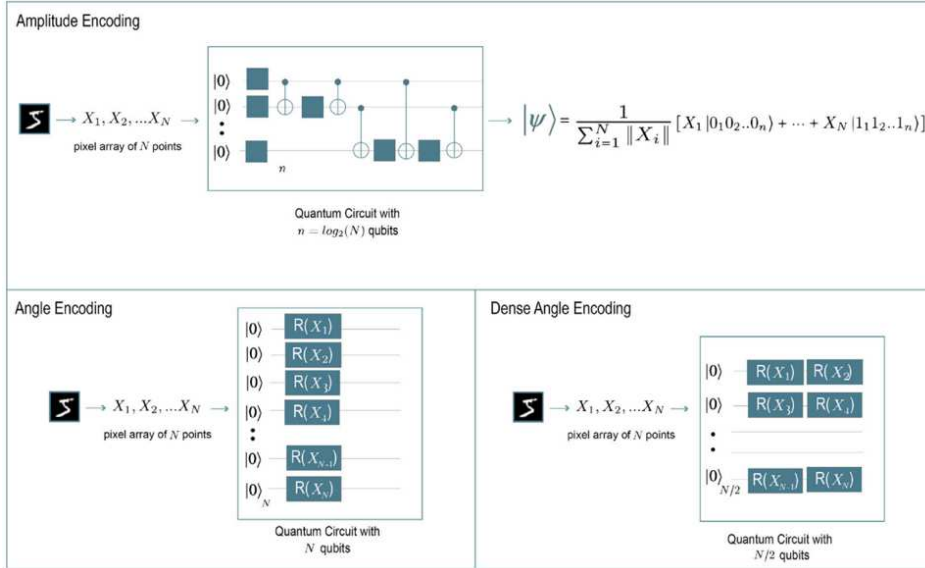


Fig. 8 Amplitude, angle, and dense angle encoding schemes[3].

3.3.4 Quantum convolution operations

The quantum convolutional layer is the central innovation of the QCNN architecture, replacing classical convolutional kernels with Parameterized Quantum Circuits (PQCs). These circuits perform feature extraction using quantum mechanical operations.

Parameterized quantum circuits (PQCs). The convolution operation uses trainable quantum gates with tunable parameters $\theta = \{\theta_1, \theta_2, \dots, \theta_m\}$. The quantum convolution operation itself can be expressed as

$$U(\theta)|\psi_{\text{in}}\rangle = |\psi_{\text{out}}\rangle, \quad (7)$$

where $U(\theta)$ is a parameterized unitary transform applied to the input quantum state $|\psi_{\text{in}}\rangle$. In their work, Hur et al. [37] demonstrated that such PQCs can achieve excellent classification accuracy despite using significantly fewer parameters compared to classical CNN models. The effectiveness of learning these parameters using non-gradient methods such as evolutionary strategies was investigated in the work of Friedrich and Maziero [17].

Figure 9 presents a collection of parameterized quantum circuits that serve as the core components of convolutional layers within Quantum Convolutional Neural Networks (QCNNs) [6]. Each subfigure (a–f) depicts a different circuit configuration built from combinations of single-qubit and two-qubit quantum gates, where $R_i(\theta)$ denotes a rotation by an angle θ around the axis $i \in \{x, y, z\}$ by an angle θ on the Bloch sphere. These circuits have been selected or adapted based on their known expressibility, entangling capabilities, and prior use in quantum machine learning tasks.

Circuit 1 (Figure 9a) corresponds to the ansatz commonly used in Tree Tensor Networks (TTN), providing a simple yet structured approach to entanglement. Circuits 2–5 (Figure 9b–e) are modified here for two-qubit systems. These circuits offer a balance between expressiveness and entanglement, and are particularly useful in designing translation-invariant convolutional layers. Circuit 6 (Figure 9f) is inspired by the two-qubit variational quantum entangler (VQE), which is capable of implementing arbitrary $\text{SO}(4)$ operations. Its structure, when arranged linearly over qubit pairs, mirrors the topology of convolutional filters, making it a suitable and expressive candidate for quantum feature extraction.

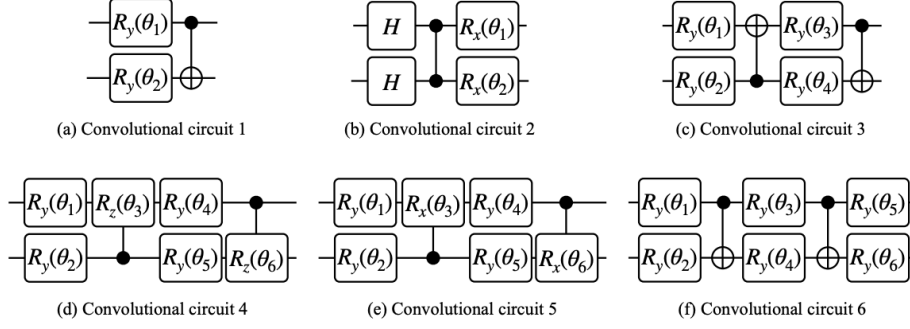


Fig. 9 Parameterized Quantum Circuits for Convolutional Layers in QCNN[6].

Quantum Filter Design. The structure of the quantum filter itself is critical to the performance. Han et al.[35] developed variational quantum convolutional neural networks (VQCNNs) based on quantum filters and fully connected layers, showing improved performance. In turn, Shi et al.[37] investigated different QCNN architectures based on variational quantum filters and demonstrated that factors such as parameter sharing and filter arrangement significantly affect the performance in multi-class scenarios. Quantum filters typically use steerable rotational gates and obfuscation operations to create trainable feature detectors that can capture nonlinear and complex correlations in data.

The quantum field is formed by embedding a small portion of the input data, in this case a 2x2 rectangle is used for parameterized rotations (R) on the ground state qubits. The machine performs a quantum computation associated with a unitary (U). A variational quantum circuit, a randomized circuit, produces a unitary. Finally, the quantum system is computed to produce a list of traditional strategic growth. The effects of the computation can be post-processed in the traditional way, but for convenience, only the raw perceptual values are used. By repeating the process in different areas, all the data are scanned, resulting in an output image that is structured as a channel with multiple inputs, as shown in Figures 10 [12].

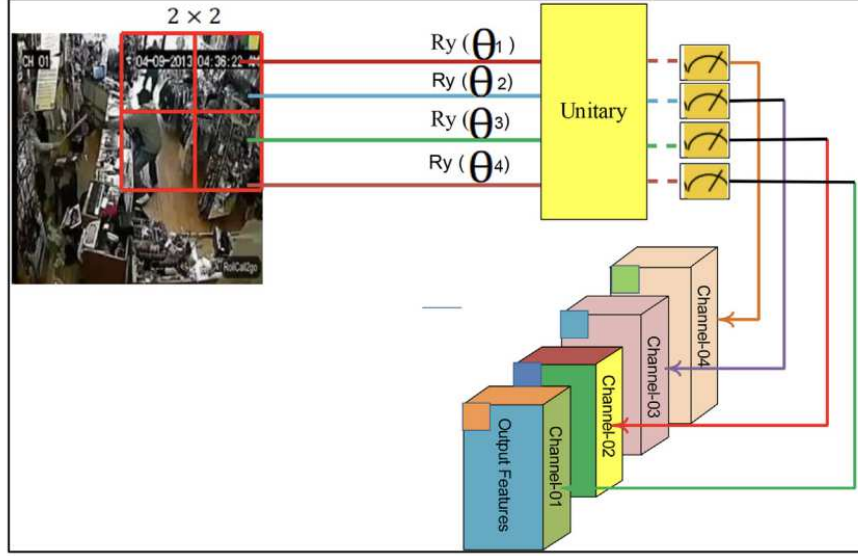


Fig. 10 Quantum image generation[12].

Figure 11 illustrates the quantum convolutional layer architecture used for image data processing. The input image, denoted as G , consists of pixel values arranged in a two-dimensional matrix, where each pixel is represented by $G_{j,k}$. These pixel values are normalized and encoded into a pure quantum state $|g\rangle$, which is constructed using a number of qubits proportional to the logarithm of the total number of pixels.

The convolution operation transforms the input image G into an output image H by applying a spatial filter represented by matrix X . This filter interacts with the quantum-encoded input state through a linear filtering operator V , which is constructed from submatrices related to the filter mask X . The operators V and its variants enable the convolution process within the quantum framework, allowing the image features to be extracted and transformed while maintaining the quantum representation.

This approach leverages the unitary evolution of quantum states to perform convolution, in contrast to classical matrix multiplication methods. The process results in an output quantum state $|h\rangle$, which contains the convolved image information. The design efficiently utilizes a limited number of qubits and energy levels, independent of the input image size, thus demonstrating how quantum operations can simulate classical convolutional filtering in computer vision tasks [14].

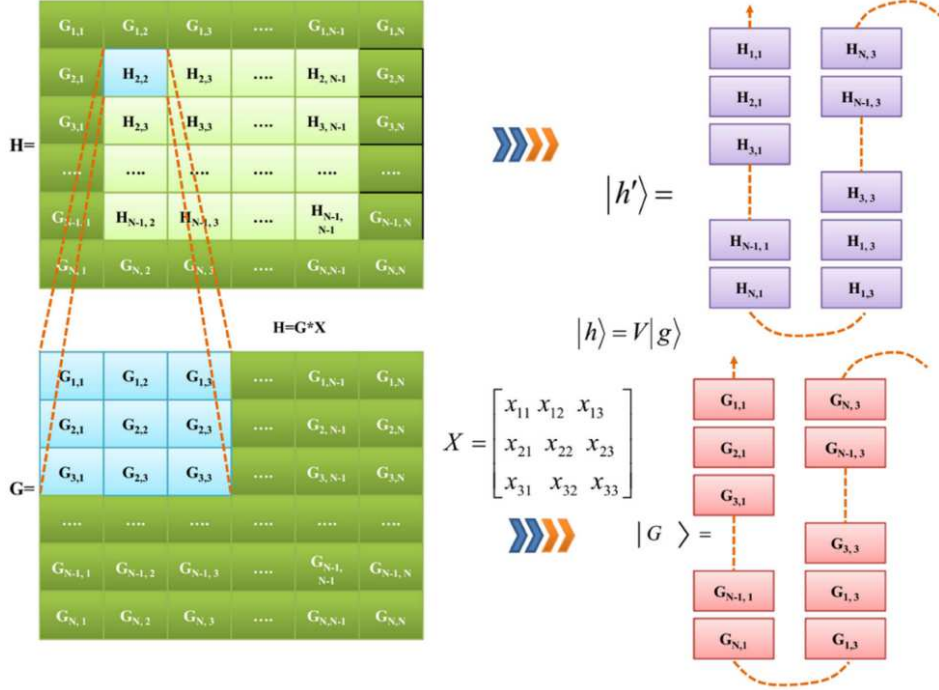


Fig. 11 Comparing quantum convolutional process with standard convolution operation [14]

3.4 Hybrid architectures (RQ4)

In the vast majority of studies (more than 90%), QCNN is integrated with classical layers to form hybrid quantum-classical models [31, 33, 36].

3.4.1 QCNN architecture

At its core, QCNN is implemented in several stages, including data encoding, one or more quantum convolution layers, and measurement. The original classical data (e.g., image pixels) is transformed into a quantum representation, this state is then passed through a trainable parameterized quantum circuit, and the qubits are measured at the output to obtain classical features or the classification result itself. Detailed diagrams illustrating this process can be found in Hur et al.[6] and Chen et al.[16].

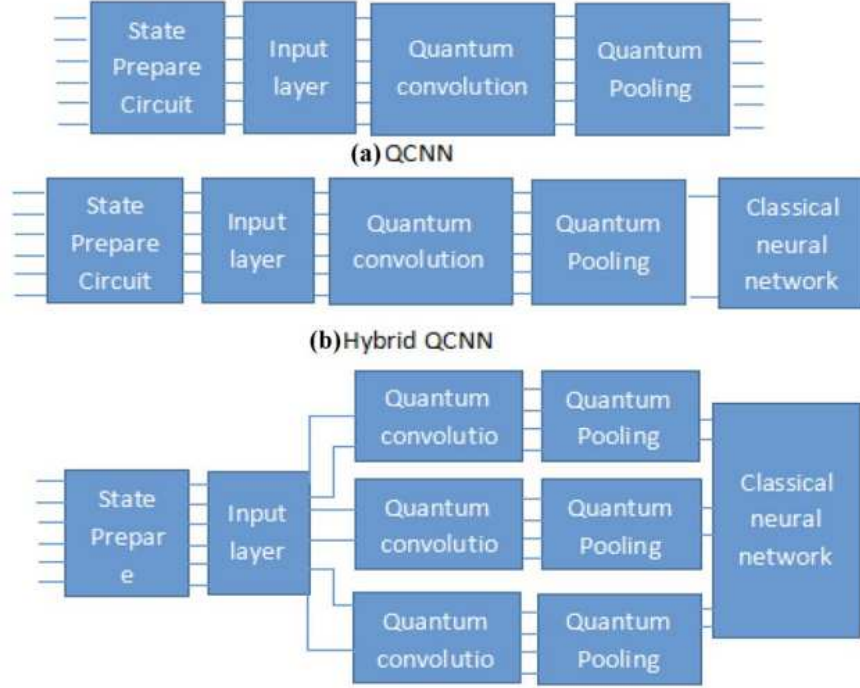


Fig. 12 QCNN and hybrid QCNNs with classical architecture [16].

The main feature of QCNN is the use of quantum filters and entanglement operations instead of traditional convolutional kernels. This concept, first called a “quantum convolutional” (quanvolutional) layer, was proposed and demonstrated in the seminal work of Henderson et al.[5]. Unlike classical filters, quantum filters can use superposition and entanglement to extract complex correlations from data that may be inaccessible to classical methods.

Figure 13 illustrates the architectural flexibility and structure of a quantum convolutional neural network based on the use of quanvolutional layers. Subfigure(A) shows an example of a complete neural network architecture, including one quanvolutional layer with three quantum filters, followed by a pooling layer, a classical convolutional layer with six filters, another pooling layer, and two fully connected layers, where the final layer is responsible for classification. This structure is identical in form to a classical CNN and can be easily modified; for example, a classical convolutional layer can be replaced by a quanvolutional layer with the same number of quantum filters, and vice versa. Subfigure (B) illustrates the operation of a single quanvolutional filter. The input image fragment is encoded into a quantum state, processed through a random quantum circuit, and subsequently measured and decoded into a scalar value, forming an output feature analogous to the activation produced by a classical convolution. This process is denoted as $Q(u_x, e, q, d)$, where u_x represents the input image fragment, e is the quantum encoding function, q denotes the quantum circuit, and d is the decoding

function that maps the measured quantum state to a classical value. Consequently, Q represents the complete transformation of classical data using the quantum filter [5].

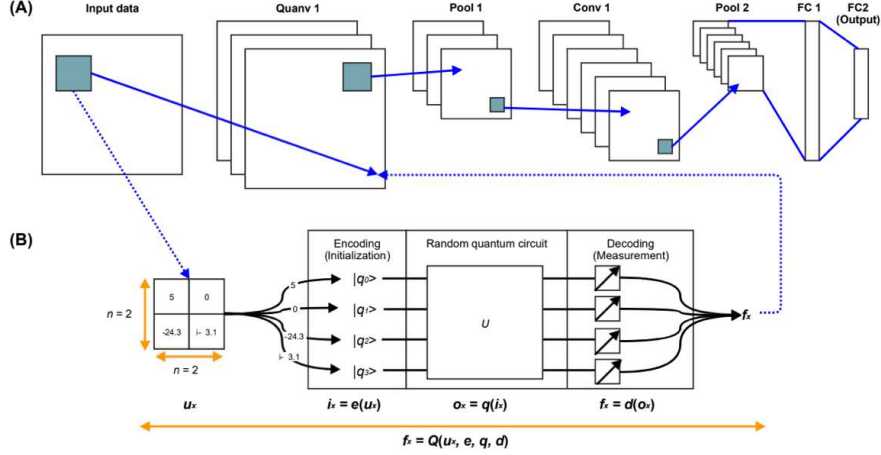


Fig. 13 A simple example of a quanvolutional layer in a full network stack [5].

Modern architectures take this idea further, for example by introducing “interaction layers” that exploit three-qubit interactions to improve performance[7], opening up new possibilities in recognizing complex patterns.

Figure 14 illustrates the structure of a quantum convolutional layer composed of repeating parameterized quantum blocks called ansatzes (green rectangles). Each block performs a two-qubit interaction applied to neighboring qubits in a translation-invariant manner, preserving structure under qubit shifts. The first column connects each qubit to its right neighbor, while the second connects to the left, ensuring full local coverage. The outer ansatzes in the second column also connect the first and last qubits, forming a closed-loop structure. All ansatzes in a layer share identical parameters, reducing the number of trainable weights. The blue rectangles in the Pooling Layer denote quantum pooling ansatzes, designed to reduce qubit count while preserving essential information. These use controlled rotation gates and Pauli-X operations, and like convolutional blocks, are parameter-shared across the layer[7].

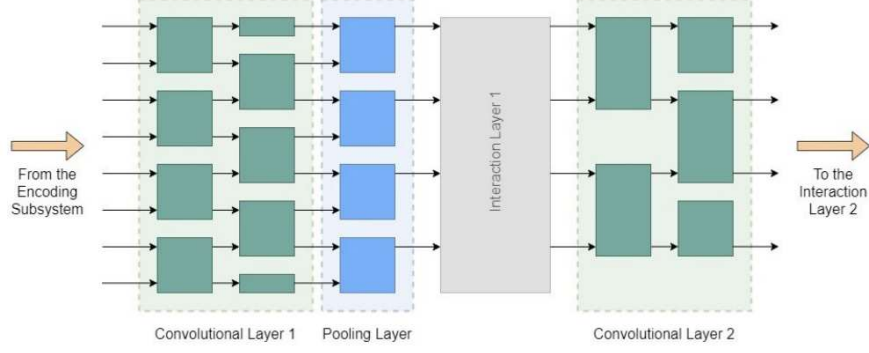


Fig. 14 The Convolutional Layers comprises ansatzes, all of which use two-qubit interactions [7].

This figure illustrates the full architecture and operational flow of the proposed HQNN model[32]. In panel (a), the overall algorithmic pipeline is depicted. It begins with an average pooling downsampling stage that reduces the dimensionality of the classical input data, followed by a quantum processing module and a classical post-processing module. The quantum part starts with an encoding circuit, where each classical input value x_i is nonlinearly mapped using the function $f(x_i) = \arcsin(x_i)$ and applied as the rotation angle in a gate $R_y(f(x_i))$. This encoding circuit transforms the classical data vector into a quantum state using a real-amplitude mapping, thereby enabling the embedding of information into the amplitudes of qubits.

After encoding, the parameterized quantum circuit (PQC) performs a series of entangling operations and unitary transformations on the encoded state. The unitary gate $U_i(\theta_i) = \exp(-i\theta_i Z_j Z_k)$, shown in panel (b), is implemented using a standard decomposition into two CNOT gates and a single Z -rotation gate with an angle $2\theta_i$, enabling efficient two-qubit entanglement within the PQC structure.

Panel (c) presents a detailed breakdown of the quantum subsystem. It consists of three parts: the encoding circuit on the left, where each qubit is prepared using a R_y gate parameterized by the encoded classical feature; the middle section showing a single layer of a ladder-structured PQC, requiring n_l trainable parameters; and the rightmost part, where quantum measurements are performed on each qubit to compute the expectation values of observables. These expectation values are passed to a fully connected classical neural network with a softmax output layer, which produces the final classification result. The entire system is trained end-to-end, with a classical optimizer updating both quantum circuit parameters and classical weights to minimize prediction loss.

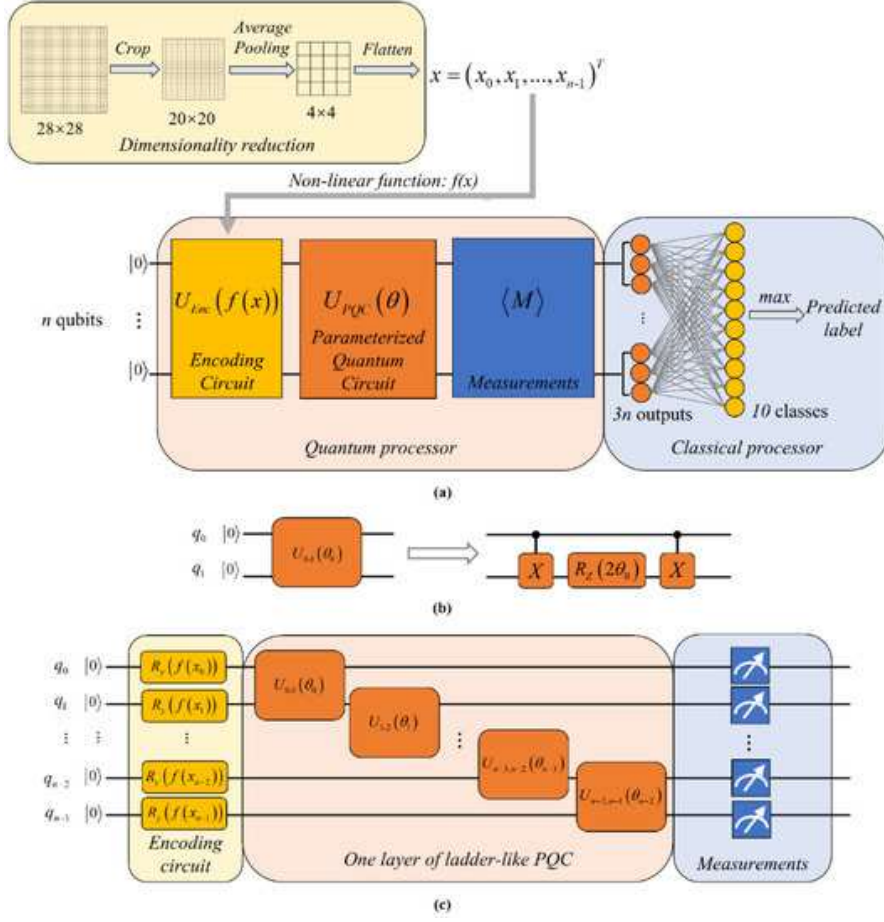


Fig. 15 Architecture of the Hybrid Quantum Neural Network [32].

This approach is the de facto standard for the NISQ era, as it allows to bypass the limitations of modern quantum processors. The classical part, often in the form of a pre-trained CNN (e.g. ResNet), pre-processes the data: reduces the dimensionality of the image and extracts low-level features. These compressed features are then fed to the quantum part for deeper analysis and extraction of complex features[15].

This diagram presents a hybrid framework that integrates classical and quantum neural network layers. Classical layers consist of standard architectures such as convolutional and fully connected layers, while quantum layers incorporate parameterized quantum circuits to enhance expressiveness. The input vector X_i is processed through the classical network, producing hidden states h_i , which are subsequently used as rotation angles $R(h_i)$ in quantum gates. The final output y is generated through a measurement step following the quantum operations[36].

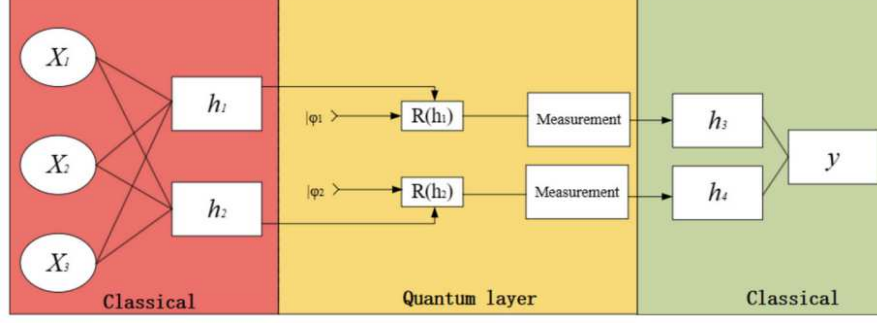


Fig. 16 Hybrid quantum neural network framework[36].

This approach is particularly effective in medical problems, where training data is often limited and accuracy requirements are high. Examples include hybrid QCNNs for detecting stenosis in angiograms[9] or for classifying X-ray images[10]. Quantum-enhanced transfer learning and ensemble methods are also actively developed, where quantum layers are added to already trained classical models or several quantum models are combined to improve the robustness and accuracy of predictions[40]. The effectiveness of hybrid models has also been demonstrated in natural language processing problems, such as sentiment analysis[38].

3.5 Quantum Optimization Algorithms (RQ5)

Quantum optimization algorithms provide specialized tools for addressing complex optimization problems inherent in machine learning tasks, particularly those involving combinatorial challenges and hyperparameter optimization.

Quantum-Enhanced Metaheuristic Algorithms. Quantum Genetic Algorithms have been successfully integrated with quantum machine learning (QML) to address optimization challenges in circuit design, feature mapping, and model training. Phalak et al. (2023) proposed a GA-based approach to optimize quantum embeddings (QEs) for quantum neural networks (QNNs), framing the search for optimal feature-to-qubit mappings as a combinatorial problem. Figure 17 illustrates a workflow in which a genetic algorithm (GA) is used to optimize the way classical features are encoded into a quantum machine learning (QML) circuit by finding their optimal permutation. The process begins in step 1 by forming an initial population, where each individual, labeled E_1 through E_k , represents a unique permutation (order) of the features to be mapped to qubits. In step 2, this population is iteratively improved by the genetic algorithm. This process involves a crossover operation (a), where parent permutations are combined to create a new generation of offspring (E'), and a mutation operation (b), where two features are randomly swapped in some individuals to maintain genetic diversity. The key step is the evaluation of the fitness function (c): for each individual E' in the new generation, a corresponding quantum circuit is constructed and simulated, where the features are encoded in the specified order. The final performance of this scheme, such as classification accuracy, becomes its fitness score. After completing a given number of iterations in step 3, the algorithm outputs a

single best individual E_{fittest} , which is the best performing feature permutation found during the entire evolutionary search.

This approach provides a balance between the learnability of the model and its generalization ability, which is especially important in the presence of noise in NISQ devices. To improve the stability of the estimates, the authors average the results of 10 runs of each embedding variant. This metric allows for efficient comparison of different ways of mapping features to qubits: for example, if one embedding provides 95% accuracy during training and 90% during testing (fitness = 92.5), while another provides 85% and 88%, respectively (fitness = 86.5), the algorithm will choose the first variant as better. During the genetic algorithm run, this feature is used to select the best individuals, perform crossover and mutation, which ultimately allows finding embedding configurations that are robust to noise and outperform random selection. Their method demonstrated superior performance compared to random selection, achieving 0.33–3.36 higher fitness scores on the MNIST and Tiny Imagenet datasets with 8.1–15% reduction in runtime, and outperformed existing methods such as quantum embedding kernels (QEK) by 1.06 times[19].

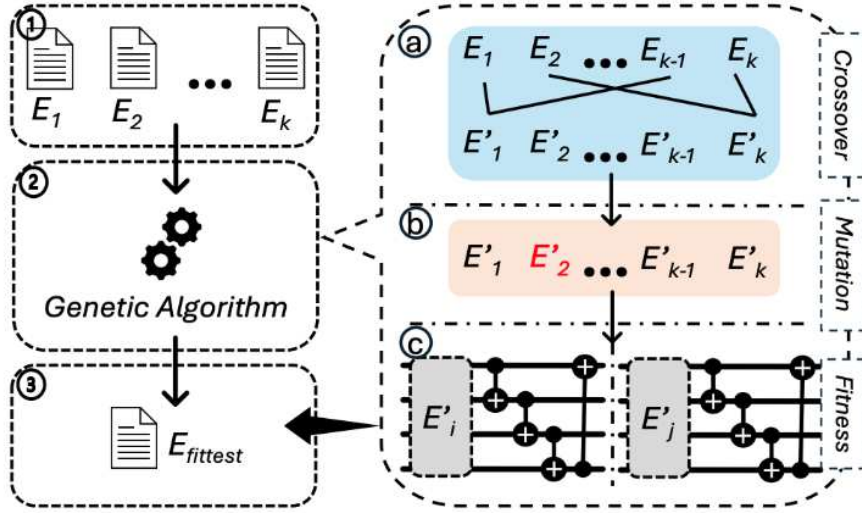


Fig. 17 QCNN’s genetic algorithm evolves embeddings by crossover, mutation, and selecting the best via quantum simulation[19].

Similarly, Gong et al. (2023) introduced QGA–QGCNN, a quantum gate circuit neural network optimized by a quantum genetic algorithm (QGA), which leveraged QGA’s global optimization capabilities to enhance convergence speed and accuracy in intrusion detection tasks, achieving lower false alarm rates and improved parallel processing for uncertain problems[20]. The authors propose a QGCNN model based on quantum rotating gates and multi-qubit operations, where the training parameters are rotation angles rather than traditional weights. In a quantum genetic algorithm (QGA), as described in Algorithm 1, individuals exist in two forms. Initially, each

individual of the population $Q(q_0)$ is represented as a quantum chromosome consisting of qubits (step 1). Due to superposition, one such chromosome can simultaneously contain information about multiple states. To assess fitness, this quantum chromosome is “observed” (step 4), causing it to collapse into a classical binary individual $P(t)$. A fitness function is then calculated (step 5), which estimates how “good” each of these classical individuals $P(t)$ is at solving the target problem.

The key evolutionary step (step 6) is the update of the initial quantum chromosomes. Instead of traditional crossover and mutation, QGA uses a quantum rotational gate $R(\theta)$. The rotation angle (θ) for each qubit is determined based on a comparison of the fitness of the current individual with the best individual found so far (step 7). If the fitness of the current individual $f(x_i)$ is higher than that of the best $f(b)$, then the probability amplitudes $[\alpha, \beta]$ of its qubits are changed to increase the probability of its reappearance. If the “elite” individual is better, the amplitudes are shifted towards it. This process allows for efficient population steer toward the optimal solution by using information about the best individual to adjust the entire population at the quantum level. The entire cycle (steps 3-9) is repeated until the stopping condition is met, which allows the algorithm to maintain population diversity while quickly converging to the goal.

Algorithm 1 QGA procedure algorithm[20]

- 1: Initialize the initial population $Q(q_0) = [q_1^0 \ q_2^0 \ q_3^0 \ \dots \ q_n^0]$ of individuals with population size n
- 2: $t \leftarrow 0$ (t is the index of generation)
- 3: **repeat**
- 4: Make the individual of binary string $P(t) = [\alpha_1^t \ \alpha_2^t \ \alpha_3^t \ \dots \ \alpha_n^t]$ by observing q_i^t
- 5: Evaluate $P(t)$ by calculating fitness of every individual
- 6: Update $P(t)$ with quantum rotation gate $R(\theta)$, and perform

$$[\alpha', \beta']^T = R(\theta_i)[\alpha_i, \beta_i]$$

for each qubit of each quantum chromosome

- 7: Record the best individual and fitness
 - 8: $t \leftarrow t + 1$
 - 9: **until** termination condition
-

For circuit optimization, Sünkel et al. (2023) developed GA4QCO, a framework employing GAs to automate quantum circuit design, addressing challenges such as barren plateaus and trainability while prioritizing low-depth circuits for practical NISQ-era applications[21]. The authors propose a flexible system where quantum circuits are encoded as individuals in a genetic algorithm, and the user can specify target properties through custom fitness functions, be it the accuracy of reproducing a given quantum state, the degree of entanglement, or efficiency for machine learning. The approach is based on a flexible system of user-defined fitness functions: (1) a fidelity function that maximizes the correspondence between the target and obtained states;

(2) an entanglement function that estimates the von Neumann entropy for generating maximally entangled states; (3) machine learning functions that optimize the accuracy of classification/regression. Key features include support for various crossover operators (single-point, multi-point, block) and mutations (gate replacement, parameter change), which allows one to efficiently explore the space of possible circuits. Experiments demonstrate that the approach achieves high accuracy (up to 0.994 fidelity) even when using a limited set of gates, significantly outperforming random search. Particularly valuable is the focus on practical application in the context of NISQ devices, where both computational efficiency and noise robustness are important.

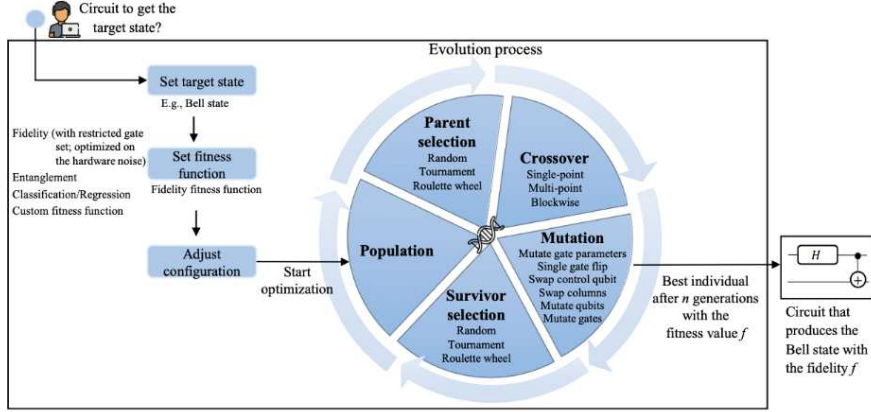


Fig. 18 Optimization of quantum circuits in the GA4QCO framework [21].

In the context of training quantum neural networks, Choe et al. (2023) demonstrated that a quantum genetic algorithm (QGA) could significantly improve quantum backpropagation (QBP) networks by mitigating local stagnation, resulting in faster convergence (up to 30% reduced training time) and higher correction rates compared to classical backpropagation (CBP)[22]. The process begins with the QGA step in Figure 20 (left side of the flowchart), where an initial population is created. In this context, individuals are potential sets of initial weights for the QBP network, encoded as quantum chromosomes consisting of qubits. These quantum individuals are first measured to obtain their classical representation, after which a fitness function is calculated for each of them. This evaluation step is key, since the fitness value of one individual is determined by the final error of the QBP neural network that was initialized and fully trained using exactly this set of weights. The algorithm keeps the best individual found, following elitism theory, to ensure that the optimal solution is not lost. If the stopping condition is not met, the population undergoes evolution. Instead of traditional operators, QGA uses quantum analogues: quantum rotation changes the probability amplitudes of qubits, steering the population towards the best solution, while quantum crossover, described as "quantum interference crossover", effectively mixes information from all individuals, and quantum mutation, implemented as "a simple inversion of two probability amplitudes", maintains population diversity. At

the end of the QGA stage, the best found set of parameters is passed to the second stage - to the QBP neural network (right side of the diagram). Here, instead of random initialization, the network starts with already optimized weights and then runs a standard backpropagation loop for local "fine-tuning" and convergence to the final solution.

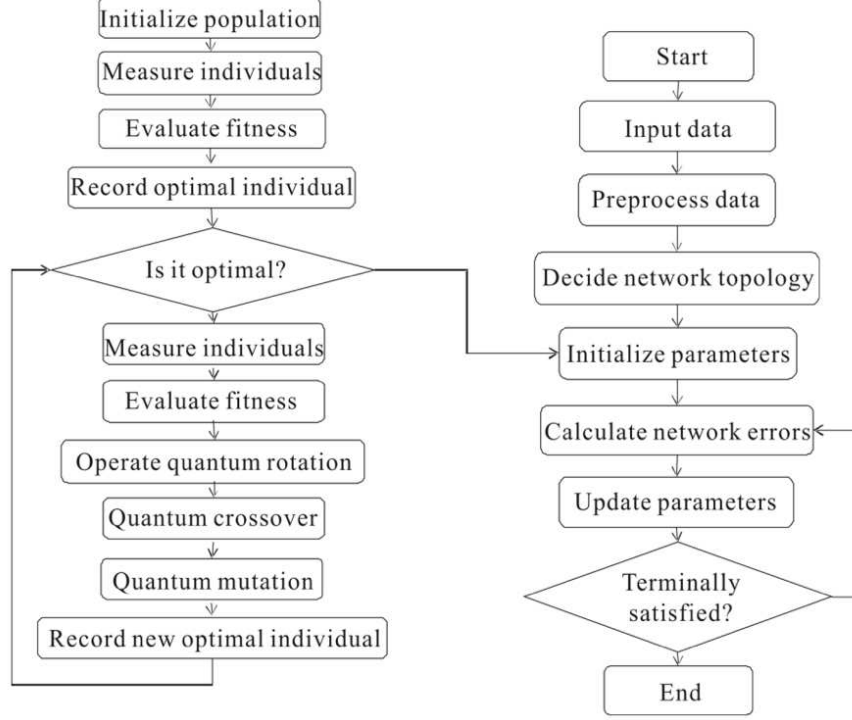


Fig. 19 Quantum backpropagation algorithm optimized using a quantum genetic algorithm [22].

These studies collectively highlight GAs as a robust tool for enhancing QML scalability, efficiency, and performance, particularly in scenarios requiring adaptive parameter optimization or resilient circuit architectures. Key advantages include reduced computational overhead, improved solution quality, and applicability to diverse QML paradigms, from embeddings to hybrid classical-quantum networks. Future work may explore hybrid GA-quantum optimization for large-scale problems and error-resilient circuit synthesis.

This comprehensive analysis of quantum machine learning models reveals a rapidly evolving field with diverse architectural approaches, each offering distinct advantages for specific image classification challenges[4, 38]. The integration of quantum computing principles with classical machine learning paradigms continues to yield innovative solutions that push the boundaries of computational efficiency and classification accuracy[24].

The specific implementation process is shown in [25]. Firstly, the primary generation of quantum gate chromosomes based on the tree structure is generated from the original quantum circuit, and the paternal chromosome is generated by the game wheel method. Then, the hybridization based on random multi-point crossover and the mutation based on the heterogeneous tree structure are applied to generate a new generation of chromosomes. Finally, in Section II .C, based on the fitting data of the quantum computation evaluation method, the two best quantum circuits are selected as the new input cycle, and as long as the generated quantum circuit computation remains unchanged for three generations, it is the optimal output quantum circuit [25].

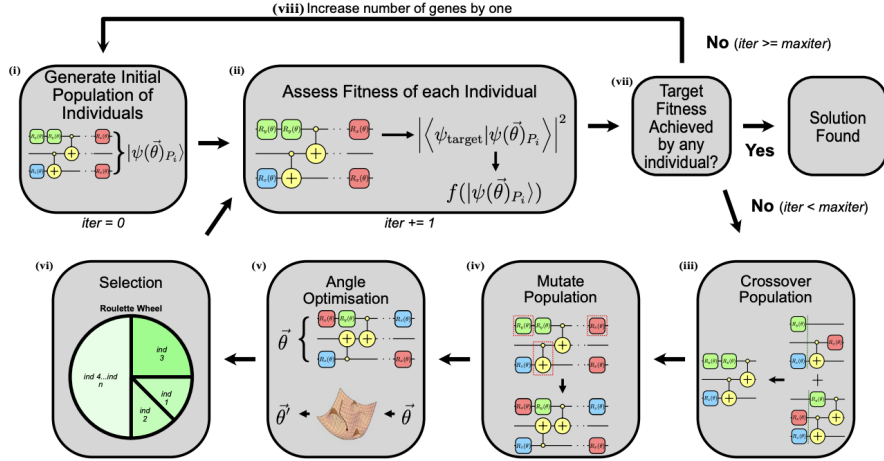


Fig. 20 Overview of the GASP approach [39].

Figure 20 illustrates the GASP (Genetic Algorithm for State Preparation) framework, where the objective is to evolve quantum circuits capable of preparing a specific target quantum state. In this approach, each individual in the population represents a quantum circuit, and each gene within the circuit corresponds to a single quantum logic gate. The process begins by generating an initial population P , where each circuit P_i consists of a predefined number of qubits and gates. The fitness of each individual is evaluated by comparing the quantum state generated by the circuit with the desired target state, thereby measuring the degree of alignment. Crossover is then applied to combine genetic information between circuits, followed by mutation with a fixed probability (typically $p = 5\%$), which introduces new quantum gates or modifies existing ones. After mutation, each individual undergoes classical optimization to fine-tune the parameter set θ within the range $[0, 2\pi]$, improving its fitness. Selection of individuals for the next generation is performed using roulette wheel selection, which favors circuits with higher fitness values. This evolutionary cycle is repeated until either a target fitness level is achieved or the number of generations exceeds a predefined threshold without improvement. If no progress is observed after a specified number of iterations ($maxiter$), the algorithm increases the number of genes (i.e.,

quantum gates) by one and restarts the evolutionary process. This adaptive mechanism enables the progressive construction of more expressive quantum circuits capable of approximating complex quantum states [39].

3.6 Performance Evaluation Metrics for Optimization (RQ6)

To comprehensively evaluate the performance of quantum circuit optimization in QCNN (RQ6), the selected studies use a two-tier approach that includes both metrics that evaluate the physical efficiency of the quantum circuit itself and metrics that characterize the overall performance of the entire machine learning model.

The first tier of evaluation is directly related to the limitations of the NISQ era and aims to quantify the success of the optimization. The key metrics here are circuit depth and gate count, with a special focus on the number of expensive two-qubit entanglement gates. Reducing these metrics is critical to minimize decoherence-induced errors and to ensure that the circuit can be executed on real quantum hardware. Studies focused on structural optimization and circuit compilation, such as [24, 27], as well as the development of specialized frameworks, such as GA4QCO [21], primarily focus on these physical metrics. Another important metric is the number of parameters, since their reduction not only simplifies the scheme, but also facilitates the training process, which has been demonstrated in works devoted to efficient data encoding [8] and the general architecture of QCNN [6].

The second level of evaluation is intended to ensure that the scheme optimization has not led to the degradation of its main function - classification. For this purpose, standard machine learning metrics are used, such as accuracy, precision, recall and F1-score, which serve as the main success indicator in most applied works [14, 15, 33]. In addition, the convergence rate and the total training time are analyzed to evaluate the effectiveness of the training process itself, especially when it is optimized using metaheuristic approaches. Studies comparing evolutionary approaches with gradient methods often use these metrics to demonstrate advantages, such as avoiding “barren plateaus” or achieving faster error stabilization [17, 22, 29].

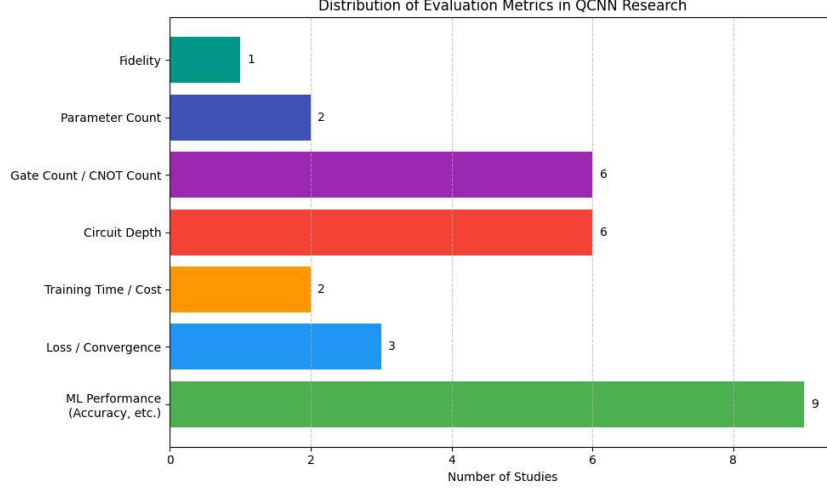


Fig. 21 Distribution of Evaluation Metrics in QCNN and circuit optimization Research

The most comprehensive and high-quality studies are trade-off analyses, explicitly or implicitly demonstrating the balance between these two levels of metrics. For example, they show how much the depth of the circuit can be reduced while minimally affecting the classification accuracy, which is a critical factor for the practical implementation of optimized QCNNs. Evaluation of different mutation techniques in genetic algorithms, as in [26], is essentially such a trade-off analysis, since different strategies lead to different trade-offs between circuit quality and model performance.

4 Conclusions

This systematic literature review provides a focused analysis of the application of heuristics and metaheuristics for quantum circuit optimization in quantum convolutional neural networks based on 40 primary studies. The findings indicate a rapid growth of this research field, especially from 2022 onwards, driven by the urgent need to adapt theoretical QCNN models to the practical limitations of NISQ-era quantum devices. Our analysis confirms that the dominant paradigm is hybrid classical-quantum architectures, where metaheuristics are applied for targeted optimization of variational quantum layers. Among these algorithms, genetic algorithms (GAs) and their variants [17, 18, 20, 21] are the most common tools used both for parameter tuning to circumvent the “barren plateau” problem [22, 29] and for structural optimization to reduce circuit depth and the number of gates [24, 27].

The main contribution of this work is to consolidate and systematize knowledge in a highly specialized but critical subfield that has not been systematically reviewed before, unlike broader topics in quantum machine learning [1, 3]. This review demonstrates that metaheuristic optimization serves as a key pragmatic bridge between the theoretical capabilities of QCNNs and the practical limitations of modern quantum computing. The results show that for complex, often non-differentiable and noisy

search spaces typical for quantum circuit design, global gradient-free search methods are currently a more robust and efficient strategy than traditional optimization methods.

Despite significant progress, our analysis reveals several critical gaps that guide future research directions. First, there is a pressing need for studies that apply and evaluate these optimization methods on more complex and large-scale datasets, going beyond MNIST and its variants, to truly test their scalability. Second, there is a need to build comprehensive, standardized frameworks to objectively compare the performance of different metaheuristics (e.g., GA vs. evolutionary strategies or swarm intelligence) on specific optimization problems. Finally, developing hardware-aware optimization algorithms that take into account the specific topology and noise characteristics of target quantum devices remains a vital research area. In conclusion, optimizing quantum circuits using heuristics and metaheuristics is not just an incremental improvement, but a fundamental technology that ensures the viability of the field of quantum computer vision. It transforms QCNNs from theoretical constructs into more practical models that can operate in the noisy and resource-constrained environment of modern quantum hardware. As quantum technologies mature, the sophisticated optimization strategies cataloged and analyzed in this review will form the basis for developing powerful, efficient, and scalable quantum AI systems.

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